

Birla Institute of Technology and Science, Pilani
First Semester, 2015-16
CHE F212, Fluid Mechanics

Quiz 4(Closed Book)

09/09/15

15 min, 5 points

NAME: _____ SOLUTION _____

ID: _____

Section: _____

A flow is represented by the velocity field mentioned below. Determine if the field is

(a) a possible incompressible flow

[2.5]

and (b) irrotational

[2.5]

$$\vec{V} = (x^7 - 21x^5y^2 + 35x^3y^4 - 7xy^6)\hat{i} + (7x^6y - 35x^4y^3 + 21x^2y^5 - y^7)\hat{j}$$

Basic equations: Incompressibility

$$\frac{\partial}{\partial x}u + \frac{\partial}{\partial y}v = 0$$

Irrotationality

$$\frac{\partial}{\partial x}v - \frac{\partial}{\partial y}u = 0$$

a) $u(x,y) = x^7 - 21x^5y^2 + 35x^3y^4 - 7xy^6$

$v(x,y) = 7x^6y - 35x^4y^3 + 21x^2y^5 - y^7$

$$\frac{\partial}{\partial x}u(x,y) = 7x^6 - 105x^4y^2 + 105x^2y^4 - 7y^6$$

$$\frac{\partial}{\partial y}v(x,y) = 7x^6 - 105x^4y^2 + 105x^2y^4 - 7y^6$$

Hence $\frac{\partial}{\partial x}u + \frac{\partial}{\partial y}v \neq 0$

COMPRESSIBLE

b) $u(x,y) = x^7 - 21x^5y^2 + 35x^3y^4 - 7xy^6$

$v(x,y) = 7x^6y - 35x^4y^3 + 21x^2y^5 - y^7$

$$\frac{\partial}{\partial x}v(x,y) = 42x^5y - 140x^3y^3 + 42xy^5$$

$$\frac{\partial}{\partial y}u(x,y) = 42x^5y - 140x^3y^3 + 42xy^5$$

Hence $\frac{\partial}{\partial x}v - \frac{\partial}{\partial y}u \neq 0$

ROTATIONAL

Note that if we define $v(x,y) = -(7x^6y - 35x^4y^3 + 21x^2y^5 - y^7)$ then the flow is incompressible and irrotational!